

You may not need to use all of the materials provided.

- 2** In this experiment, you will investigate the relationship between surface area and friction as a rod is hammered into a container of rice.

(a) You are provided with two pencils labelled P and Q.

- (i)** Measure and record the distance d between the opposite flat faces of pencil P, as shown in Fig. 2.1.

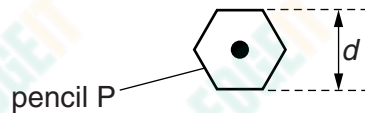


Fig. 2.1

$d =$ cm [1]

- (ii)** Measure and record the length x between the two marks on pencil P, as shown in Fig. 2.2.

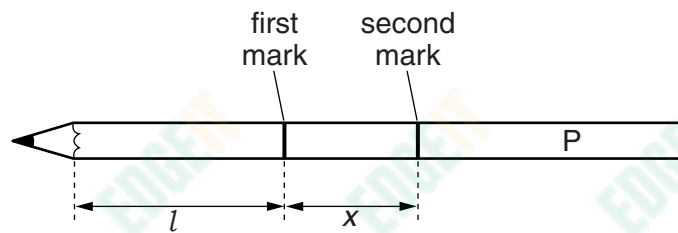


Fig. 2.2

$x =$ cm [1]

- (iii)** Measure and record the length l along the flat section of the pencil to the first mark, as shown in Fig. 2.2.

$l =$ cm [1]

(b) Estimate the percentage uncertainty in your value of l .

percentage uncertainty = [1]

- (c) Calculate the surface area A of the pencil up to the first mark, using the relationship

$$A = 2\sqrt{3}dl.$$

$A =$ [1]



- (d) (i) Push the sharp end of the pencil vertically into the centre of the container of rice until the first mark is about 1 cm above the surface of the rice, as shown in Fig. 2.3.

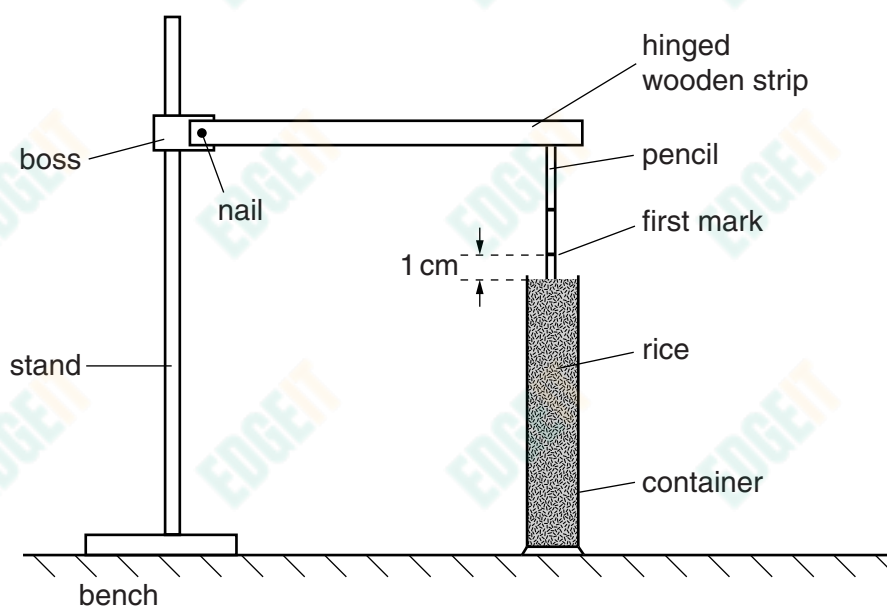


Fig. 2.3

- (ii) Position the stand and wooden strip so that the free end of the wooden strip rests on top of the pencil. Adjust the height of the nail so that the wooden strip is horizontal, as shown in Fig. 2.3.
- (iii) Raise the free end of the wooden strip by 3 cm and drop it onto the pencil. Repeat until the first mark on the pencil is level with the surface of the rice.
- (iv) Adjust the height of the nail so that the wooden strip is horizontal again.
- (v) Raise the free end of the wooden strip by 3 cm and drop it onto the pencil. Repeat until the second mark on the pencil is level with the surface of the rice, counting the number n of times the wooden strip is dropped.

$n =$ [2]



(e) Repeat (a), (c) and (d) using pencil Q.

$d =$

$x =$

$l =$ ☐

$A =$

$n =$ ☐

[2]

(f) It is suggested that the relationship between n and A is

$$n = kAx$$

where k is a constant.

(i) Using your data, calculate two values of k .

first value of $k =$

second value of $k =$

[1]

(ii) Justify the number of significant figures that you have given for your values of k .

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..... [1] ☐

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(iii) Explain whether your results in (f)(i) support the suggested relationship.

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(g) (i) Describe four sources of uncertainty or limitations of the procedure for this experiment.

1.

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2.

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3.

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4.

..... [4]

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(ii) Describe four improvements that could be made to this experiment. You may suggest the use of other apparatus or different procedures.

1.

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2.

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3.

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4.

..... [4]

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